

## Abstract

Rural electric substations coordinate protective relays that must respond to faults within microseconds, a timing error of a few hundred microseconds can trip a healthy line or fail to isolate a faulted one. IEEE C37.238-2017 governs this through a layered accuracy budget: a grandmaster clock's own time inaccuracy, plus network distribution inaccuracy, must together stay within the accuracy target of the end devices it serves, as tight as  $\pm 1\mu\text{s}$  for IRIG-B replacement applications at the substation edge.

Industrial GPS-disciplined grandmaster clocks meet this at \$30,000 per site, a cost that puts low-revenue rural substations out of reach at scale. This research tests whether a commodity Raspberry Pi 5 with GPS/PPS discipline can serve as the grandmaster in that budget at a fraction of the cost, without consuming the accuracy margin downstream devices depend on. A 7-day continuous soak on live network infrastructure measures the grandmaster's own contribution to that budget directly.

## Introduction

IEEE C37.238-2017 defines total time inaccuracy as the sum of three components: grandmaster time inaccuracy, source time inaccuracy, and distribution time inaccuracy. Each end device downstream, a protective relay, a SCADA timestamp, an IRIG-B replacement device, checks that sum against its own accuracy category, from  $\pm 100\text{ms}$  for visual indicators up to  $\pm 1\mu\text{s}$  for the most demanding substation applications.

This project's grandmaster clock is one component of that budget, not the whole chain, because the two are easy to conflate and the standard itself keeps them separate. A grandmaster that consumes little of the  $\pm 1\mu\text{s}$  total leaves real headroom for the network path and end devices it will eventually serve.

The grandmaster clock (gm) operates on the University of Idaho NILE Zero Trust network at the CIIR Coeur d'Alene facility; a fixed, network-connected deployment representative of a substation control room. A second unit, built on identical hardware and GPS/RTC HAT, operates independently as a field-portable demonstration node (fc), running live at this poster session on battery power with no network dependency. Together they represent the two deployment contexts this architecture must serve: a stationary grandmaster distributing time to substation equipment, and a self-contained unit demonstrating that the same sub-microsecond performance travels with the hardware.

## Results

### Grandmaster clock:

A 7-day continuous soak (uninterrupted multi-day run) (2026-07-01 to 2026-07-08, 37,164 samples, zero data gaps) on a GPS/PPS-disciplined grandmaster clock produced:

- Steady-state RMS (Root Mean Square) offset:  $0.2178\mu\text{s}$
- P95 offset:  $0.4897\mu\text{s}$  / P99 offset:  $0.9376\mu\text{s}$
- Two transient events: a brief antenna-acquisition window at test start and  $\sim 40$  second reference source switch traced to a DHCP reservation gap, resolved.
- A second, independently-run soak test corroborates this result at  $0.1643\mu\text{s}$  steady-state RMS, demonstrating reproducibility across separate test windows.

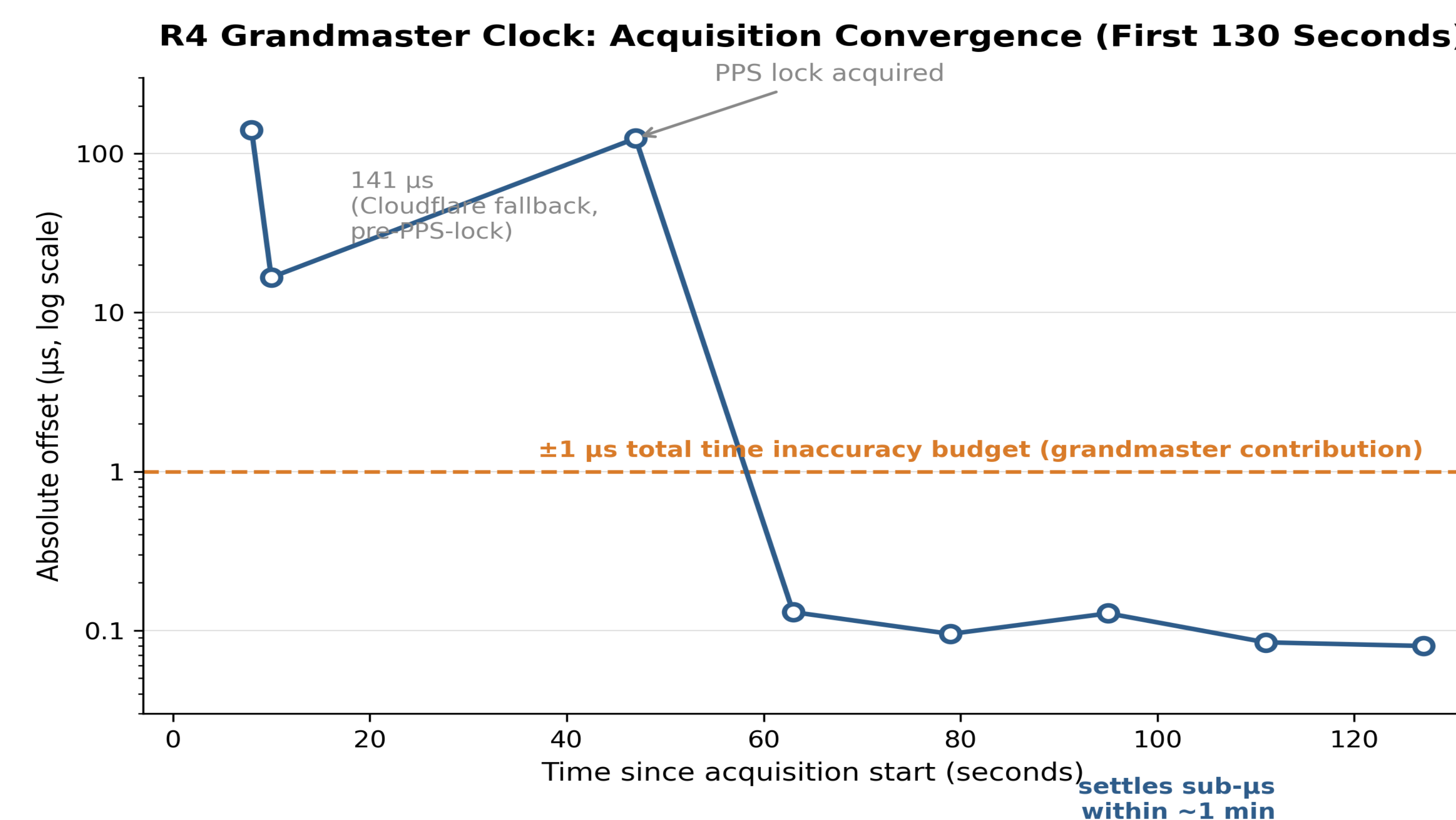


Figure 1: Data points reflect chrony's native  $\sim 16\text{s}$  polling interval; sparse early samples during Cloudflare fallback occurred at 2-8s intervals before PPS lock.

### Field-portable unit

A field-portable unit, built on identical hardware, was tested under repeated GPS-denial conditions to characterize holdover behavior.

Across five antenna-disconnect trials:

- GPS signal loss detected (PPS reach  $\rightarrow 0$ ) in 165-240s (mean 199s)
- Reacquisition completed in 195-226s (mean 214s), with 41-84 ms accumulated offset at first acquisition.
- A key methodological finding: the GPS reference reach counter stays pinned at a healthy value even during physical antenna disconnection, only the PPS (Pulse Per Second) reach counter reliably tracks real signal loss, a distinction relevant to anyone monitoring GPS-disciplined systems in the field.

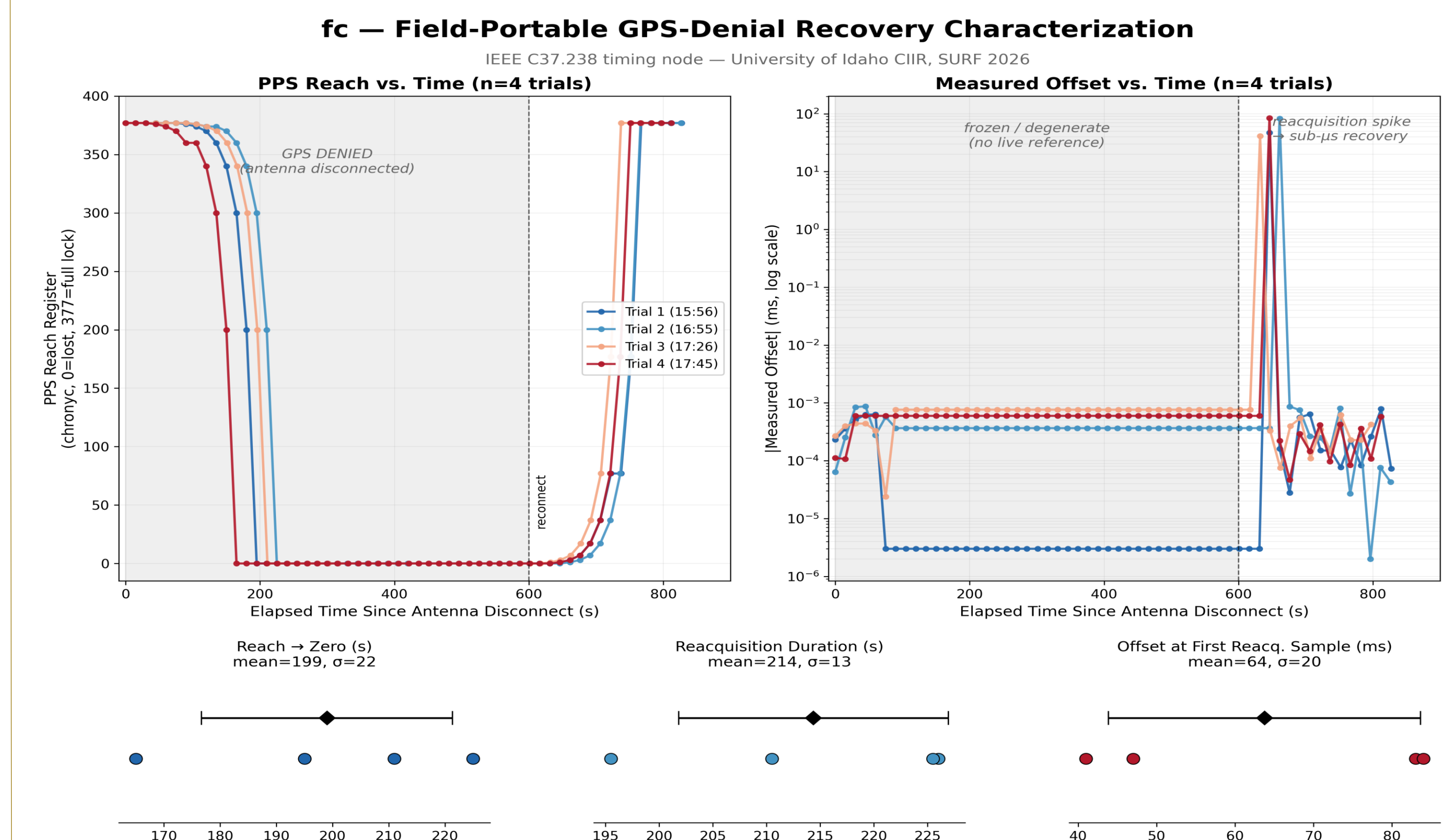


Figure 2: Pulse Per Second reach and offset across  $n=4$  trials.

## Conclusions and Future Directions

A \$280 Raspberry Pi 5 grandmaster clock contributes as little as  $0.2178\mu\text{s}$  to the total time inaccuracy budget, leaving substantial headroom within the  $\pm 1\mu\text{s}$  target that governs the most demanding substation end devices, at roughly 1/107th the cost of industrial GPS-disciplined hardware.

**Future work:** characterizing distribution time inaccuracy across a multi-hop network cascade to quantify the network's contribution to the total budget; field-deployable holdover characterization under cold-boot conditions; technical white paper development for sponsors.

## References

- [1] IEEE Standard C37.238-2017, "IEEE Standard Profile for Use of IEEE 1588 Precision Time Protocol in Power System Applications," IEEE, 2017
- [2] Chrony NTP v4.5 chrony.tuxfamily.org
- [3] J. Shovic et al., "NILE: Zero Trust Architecture for Critical Infrastructure Control Systems," University of Idaho CIIR, Technical Report CIIR-2024-03, 2024.

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